European Green Crabs and Shore Crabs: Good news for future shore crab physiology

Ryan Luvera

Abstract

Invasive species completely change ecosystems negatively impacting native species, economically important fisheries, and human ecosystem resources. Invasive species in marine environments are especially damaging due to the high dispersal events from shifting currents and high fecundity. Physiological constraints are the most important for determining this dispersal of invasive species. Salinity in the Salish Sea, USA is expected to change under future climate scenarios, potentially impacting native species physiology and biotic barriers, which can solicit invasive species like European Green Crab. Our study examined physiological responses of shore crabs (*Hemigrapsus* spp.) and European green crabs (EGC) to differing salinity conditions. We tested hypotheses that lower salinity would impact shore crab by increasing righting time, decreasing hemolymph osmolality, and elevating respiration rates. We also hypothesized that EGC would outperform shore crabs under all treatments. Here we show that EGC had no physiological change under differing salinities and shore crabs had little physiological differences under lower salinities, and possibly some physiological change under higher salinities. Neither species showed statistically significant changes in righting time or respiration with changing salinity, as indicated by overlapping error bars. Osmolality analyses revealed no significant differences between species or treatments overall, however shore crabs in the highest salinity (45 ppt) exhibited significantly higher osmolality relative to baseline. Contrary to expectations, our results demonstrate that EGC may not consistently outperform shore crabs under shifting salinity conditions. These findings suggest that both species may exhibit physiological resilience to changing salinity. Understanding these physiological coping similarities between shore crabs and EGC may influence the biotic barrier that can constrain invasion through competition for resources.

Introduction

Invasive species are one of the largest global threats. Homogenizing ecosystems by predating, competing, and spreading disease to native species, invasive species drastically increase the risk of native species extinction (Gurevitch & Padilla, 2004). Countries around the globe are losing around millions dollars each year due to invasive species presence (Marbuah et al., 2014). Due to the economic and ecological burden invasive species exhibit, understanding what controls species distribution is important to limit spread and understand ecosystem impacts.

European Green Crabs (*Carcinus maenas*, henceforth EGC) are one of the worst marine invaders globally (Yamada, 2001). Originally from the Atlantic Ocean and Baltic Sea, EGC have invaded much of the world's oceans from Australia, South America, South Africa, to both coasts of the North America (Yamada, 2001). In global ecosystems, EGC are voracious predators who compete with and completely homogenize and dominate native ecosystems (Ens et al., 2022a).

In Washington State, USA, EGC are a newer threat. First detected in 1998 on the Washington peninsula (Jamieson et al., 1998), EGC have been spreading into the larger Salish Sea which has prompted more control efforts to take place (Robinson et al., 2024). In the Salish Sea, EGC have the potential to compete with and predate on bivalve and decapod populations, destroy eel grass (*Zostera* spp.) meadows, and spread disease to native arthropods (Ens et al., 2022b).

One of the most impactful constraints to invasive species spread in marine systems is physiological factors. In fact, EGC have been predicted to invade more areas due to climate change impacting thermogeography (Compton et al., 2010). This results in invasive species within marine systems typically having wider physiological tolerances than native species (Bates et al., 2013). Wider ranges to salinity, water temperature, and oxygen deprivation are typical attributes for marine invaders, allowing them to outcompete native species when poor conditions arise (e.g. Kenworthy et al., 2018). EGC have begun traveling up the coast towards British Columbia and Alaska (Villar et al., 2026) where they are likely reaching their thermal minimum. Though EGC are thought to be euryhaline and eurythermal species allowing them to inhabit a wide range of environmental conditions, it is unclear how they may be facing biotic barriers to invasion.

Adult EGC in Washington have been found to have a diet primarily consisting of the genus *Hemigrapsus* spp. crabs (Henceforth, shore crabs)(Fisher et al., 2024). But juvenile EGC directly compete with shore crabs for space and resources on shorelines meaning shore crabs can become a biotic barrier to preventing detrimental impacts to invasion. However, this biotic barrier may fade due to climate change in the Salish sea decreasing salinity which could impact shore crab health (Walker et al., 2022). Understanding how physiological constraints impact both EGC and shore crabs has direct influence on the competitive relationship between them.

Here we present a study comparing the physiological responses of shore crabs and EGC to changing salinities. Our guiding question focused on how changing salinity fluxes seen in future projects on the Salish sea (Walker et al., 2022) may influence the physiological behavior and measurements of EGC and shore crabs. We hypothesized that lower salinities would be harder to cope with for both crabs. Extrapolating from this we hypothesized: an increased righting time at lower salinities, lower salinity would result in reduced hemolymph osmolality, and that lower salinity would result in more respiration. Finally, in all our treatments we expected that EGC would be able to outcompete shore crabs in physiological conditions.

Methods

Our experiment consisted of 3 EGC, 12 shore crabs, and 2 osmolality baseline shore crabs. Crabs were reared in ambient temperature and split equally and into salinity treatments of either 15 ppt, 35 ppt, or 45 ppt seawater for three weeks. Each week we recorded crab weight, righting time, and respiration rate. Our first and final week included hemolymph extractions to determine baseline and final osmolality.

Righting time

Righting time, or the time it takes for a crab to correct itself after being placed dorsal side down, was used as a proxy for general physiological health. A faster righting time typically correlates to a more healthy crab. We measured righting time at the beginning of our experiment, and each subsequent week.

Respiration rates

Metabolic activity was measured using a resazurin assay as a proxy for respiration rate. Crabs were placed in individual wells with resazurin, along with a blank chamber. Crabs were there left in resazurin for a total of 60 minutes, while taking measurements every 30 minutes. Respiration rate was calculated as a function of fluorescence of resazurin after 60 minutes. We used a spectrophotometer plate to quantify changes in resazurin fluorescence.

Hemolymph extractions

Hemolymph was extracted from two crabs at the beginning of our experiment to provide an average baseline osmolality. After three week in treatment conditions, we extracted the hemolymph from all crabs within treatments and compared their osmolality against our baseline to examine differences.

Data analysis

To analyze our data, we used R (version = 4.5.2). To test differences in osmolality between baseline rates and final osmolality, we used an ANOVA and two sided t-tests. To test hypothesis related to respiration rate, we used emmeans to quantify the rate of change in resazurin color change in differing salinity treatments.

Results

Our experiment had a total of 12 shore carbs (starting weight: avg=2.52g, sd =0.956g) and 3 EGC (starting weight avg = 27.42g, sd=6.38g). Our week 1 35ppt shore crabs had improper experimental setup which resulted in a complete loss, though they were replaced shortly after death.

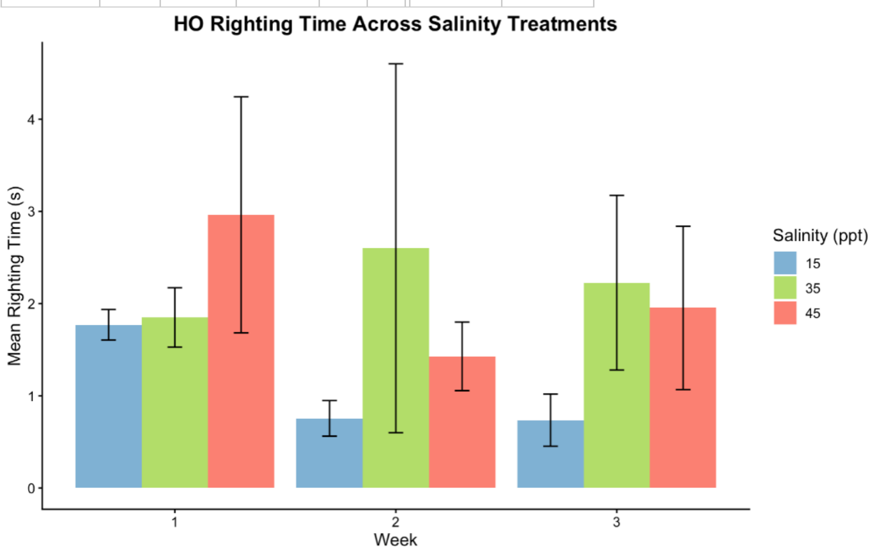
Our beginning righting times for shore crabs and EGC in all treatments (Shore crab avg = 2.19s, sd =1.5s, EGC avg = 5.05, sd=1.42s), compared to our ending righting times in all treatments (Shore crab avg = 3.86s, sd =1.45s , EGC avg = 11.18s, sd=16.4s), showed no statistical support that righting time would be higher at lower salinities (Fig.1). This conclusion was found through the fact that the error bars are overlapping for each treatment and week average highlighting the lack of statistical significance. We also found that EGC righting time typically tended to be lower in treatments, however this is a result from a single EGC replicate.

Our osmolality baseline from our two shore crab was found to be 957.5 mmol/kg. Comparing our osmolality from our final week across all treatments, EGC (avg = 1007.67mmol/kg, sd = 156.02mmol/kg) and shore crabs (avg = 1267.8mmol/kg, sd = 406.67mmol/kg) were found to have no statistical significance with a two-way ANOVA testing differences between species-treatment pairs (Fig. 3). However, when each shore crabs salinity group was independently tested against the average baseline osmolality taken in week 1 with two-sided t-tests, shore crabs in the 45ppt treatment presented an osmolality significantly higher than the baseline (p-value = 0.006785, df = 3), however shore crabs in the other salinity treatments were not significantly different than the baseline (Fig. 3).

Our resazurin treatments showed a difference from week 1 EGC respiration (avg = 3919.66, sd= 1733.46) and shore crabs (avg = 6090.66, sd= 1768.52) to week 3 EGC respiration (avg = 2475.3, sd=399.76) and shore crab respiration (avg=4684.3, sd=800.4). Our hypothesis that lower salinity would result in a higher respiration rate was not supported by our data due to overlapping error bars in our respiration means (Table 1, Fig 2).

Table 1. Estimated Marginal Means (categorical version of slope) of species respiration rate. Emmean is the slope of the respiration rate proxy. No statistical significance was found within between these means.

Treatment	Week	Species	emmean	SE	DF	lower CL	upper CL
15	1	EGC	2857	946	4	231	5483
35	1	EGC	5920	946	4	3294	8546
45	1	EGC	2982	946	4	356	5608
15	2	EGC	2367	75.2	4	2158	2576
35	2	EGC	2342	75.2	4	2134	2551
45	2	EGC	2245	75.2	4	2036	2453
15	3	EGC	2333	429	4	1141	3526
35	3	EGC	2928	429	4	1736	4120
45	3	EGC	2168	429	4	975	3360
15	1	Hemi	5953	964	4	3277	8629
35	1	Hemi	7924	964	4	5248	10600
45	1	Hemi	4395	964	4	1719	7071
15	2	Hemi	2417	144	4	2016	2817
35	2	Hemi	2296	144	4	1896	2696
45	2	Hemi	1947	144	4	1547	2348
15	3	Hemi	4739	480	4	3406	6072
35	3	Hemi	5456	480	4	4122	6789
45	3	Hemi	3858	480	4	2525	5191



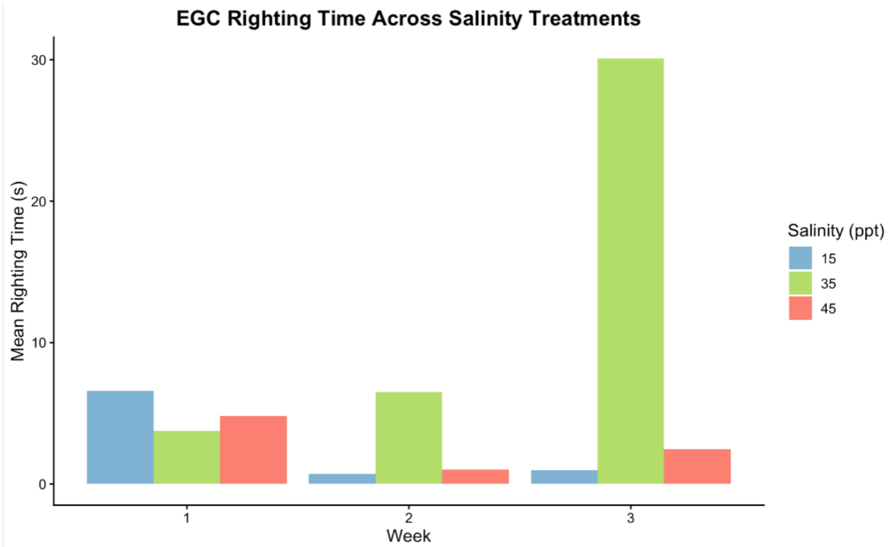


Figure 1 a, b, - Mean righting time by mean weight of European green crab (EGC) (1a) and *Hemigrapsus* (Shore crabs) (1b)

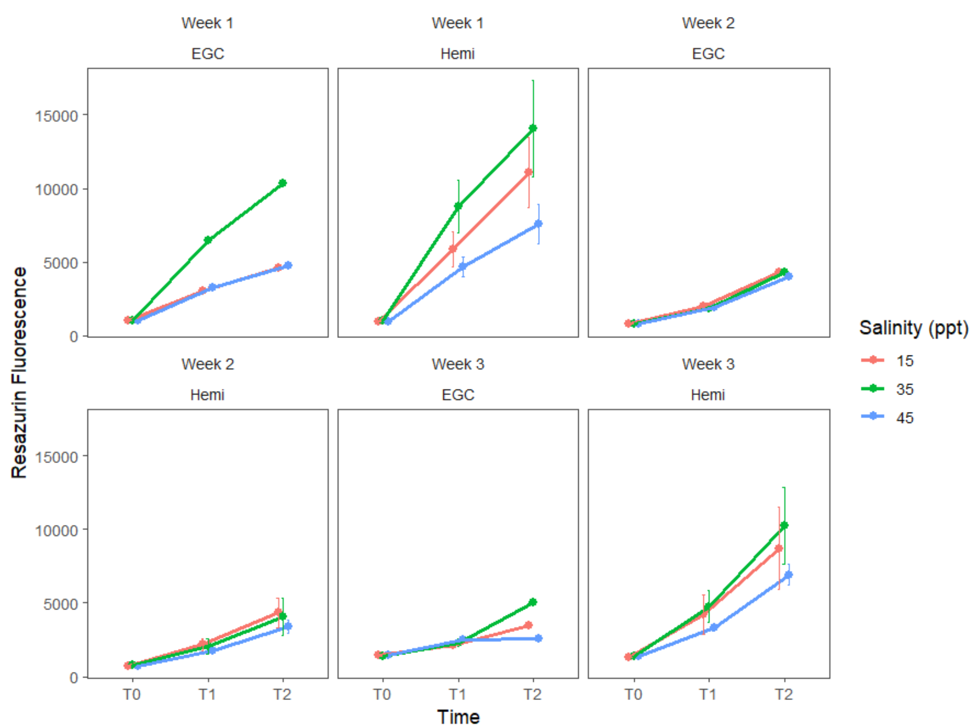


Figure 2. Hemigrapsus were submerged in fluorescence of resazurin, respiring over time. Fluorescence is a proxy for oxygen respiration. Error bars represent standard error.

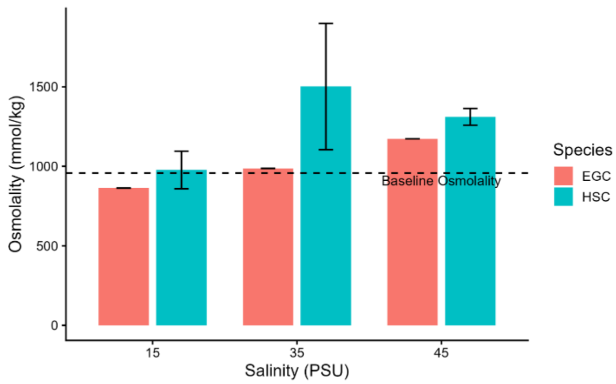


Figure 3. Osmolality (mmol/kg) of hemolymph after 3 weeks of sustained salinity treatments indicated on x-axis (15, 35, 45 ppt). Red bars indicate European Green Crab and display one crab per treatment and therefore no standard error. Blue bars indicate shore crab and display 3 crabs for 15 and 35 ppt, and 4 crabs for 45 ppt with standard error bars given those populations. Baseline osmolality measurement (dashed line) taken from 2 averaged, non-experimental shore crab before the start of the experiment.

Discussion

EGC in the Salish sea and beyond are an existential threat to the ecosystems they invade. Shore crabs may provide a biotic barrier to invading EGC by competing for resources and space. However, changing salinity conditions due to climate change may impact this competitive biotic barrier if shore crabs are not able to physiologically cope as well as EGC.

Contrary to our hypothesis that righting time would be lower for shore crabs, we found that righting time was statistically the same across treatments. We were unable to run stats on EGC due to the limited sample size, though their righting time was generally lower than shore crabs which could be an effect of larger size. Overall, this means that shore crabs righting time were not affected by differing salinity treatments. This result shows that generally physiological stressors over long periods do not determent the health of shore crabs in the manner we expected. Previous work has conferred these results with EGC having a wide salinity tolerance (Dal Pont et al., 2022) as well as shore crabs (Todd & Dehnel, 1960). One interesting incidental finding we observed was a molting of EGC during our experiment. This molting resulted in a drastic increasing in righting time to almost 30 seconds before it was able to right itself. This result shows that molting can impact EGC physiology and ability to cope which should be explored in future studies.

We found that osmolality was largely statistically the same between treatment pairs for shore crabs and EGC, contrary to our hypothesis. We actually found that the osmolality of shore crab hemolymph under salinity of 45ppt resulted in a statistically significant result higher than our baseline. The other treatments also had nonsignificant differences in osmolality across treatment in shore crabs. This means that shore crabs are likely able to cope better with lower salinity compared to higher salinity. This is a benefit for shore crabs in the Salish sea as salinities are expected to freshen the sea. Previous literature

Using our proxy for respiration rate, resazurin fluorescence, we found no significant results, contrary to our hypothesis that lower salinity would result in a lower respiration rate for shore crabs and EGC. This result is interesting as respiration rates generally are thought to increase at both high and low salinity groupings for shore crabs (Dehnel, 1960) and EGC (Taylor, 1977). Without a baseline osmolality for EGC, we weren't able to compare changes in salinity to a beginning measurement. Overall, our finding suggests that respiration rates are the same for EGC and shore crabs under differing salinity conditions, which is beneficial for shore crabs competing with EGC.

Our study found evidence that shore crabs were able to generally physiologically cope with salinity compared to EGC. The only negative physiological impact we observed was from higher salinity treatments having negative impacts on shore crab osmolality. Overall, these result are not super surprising due to the euryhaline nature of these crabs, but it does show that the biotic resistance from competition between native shore crabs and EGC may still be possible under lower salinity conditions.

Despite our positive results, this study should be treated with utmost caution however due to important caveats. First, due to a small EGC sample size we were not able to draw substantial statistical conclusions from our data about EGC. This is very important as understanding the variation in EGC physiology under differing salinity is just as important as shore crab physiology. Secondly, a large portion of crabs died early in our study period due to an ammonia spike from a natural death. This limited our sample size, and reduced the amount of data for our analysis as we decided to go forward with averaging measurements over treatment rather than examining individual effects. Finally, an EGC molted which likely affected respiration rates due to the increased physiological and metabolic demands.

Our study highlights the lack of physiological change in shore crabs under differing environmental conditions. This has implications of an intact biotic barrier as the Salish sea continues to freshen due to climate change. Understanding the competitive behavior between shore crabs and juvenile EGC in experimental conditions is the next step to disentangling the possibility of a biotic barrier. Future studies should also include examining the impacts of differing salinity on the competitive behavior of both crabs. Understanding the diet of EGC in differing life stages will also be very important for further understanding the biotic barrier. Because adult EGC are often found with shore crabs in their stomachs (Fisher et al., 2024), there may be a positive feedback loop towards surviving colonizing adults breaking down the competition between shore crabs and EGC by predation. Potentially seeding shore crabs in places where EGC and present may dampen the impact of juvenile EGC growth via competition. EGC pose an existential threat to the Pacific ocean, impacting vital fisheries and ecological functions. Discovering biotic barriers to dispersal is crucial for reducing the impact of EGC, and ultimately saving our native ecosystems from homogenization and economic disparity. Our study showcases one way in which a native crab may be able to contest EGC, but it is up to future research to provide solid support for a biotic barrier for EGC dispersal.

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